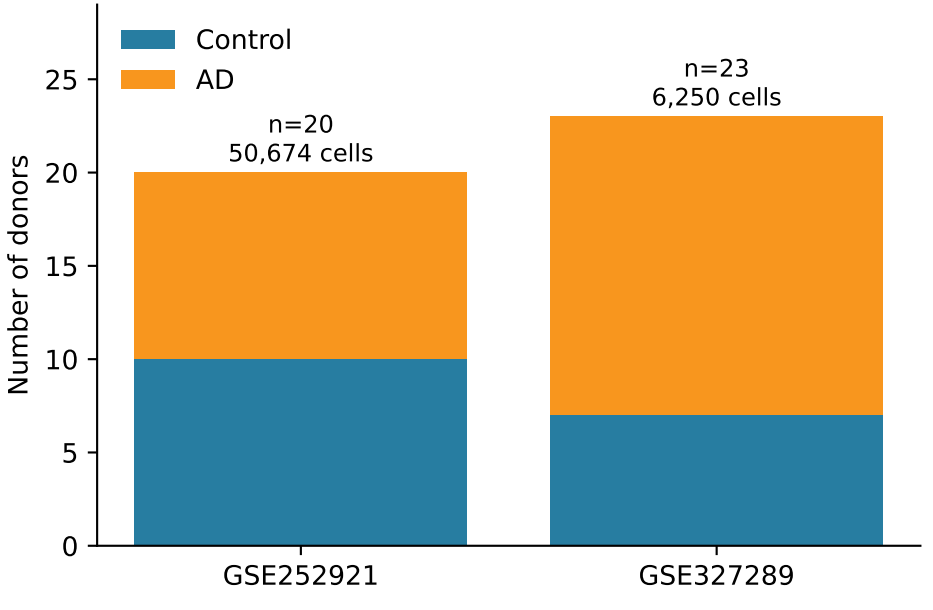
a Study design and analysis workflow

GSE252921 — Discovery
20 donors | 50,674 endothelial cells

GSE327289 — Validation
23 donors | 6,250 endothelial cells

Harmonized cross-dataset analysis
10,901 shared genes | 1,168 pathways

b Cohort composition



c Donor-aware statistical analysis

Discovery model
Disease + age + sex + batch

Validation model
Diagnosis + age + sex

Sensitivity analyses
Low-cell donor exclusion
Endothelial composition adjustment
Capillary endothelial subset

Donor independence across datasets:
Not established